

TRACE User Guide

None

TRACE Development Team

None

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1. Welcome to TRACE

TRACE (Triggered Radiotherapy Adaptive Control Engine) is a dedicated software suite designed to control and verify dynamic anthropomorphic thorax phantoms used in 4D Radiotherapy (4DRT), motion management research, and image-guided radiation therapy (IGRT).

As described in *Lustermans et al. (2024)*, TRACE interfaces with multi-axis motion hardware and triggering electronics to simulate realistic patient breathing motions and evaluate 4DCT and radiotherapy treatment delivery.

1.1 Key Features

- **Analytical & Patient-Derived Motion Profiles:**
 - Generate synthetic respiratory waveforms (sinusoidal, $\cos^4$), custom analytical functions).
 - Import and process patient-derived respiratory curves from 4DCT/RPM clinical logs.
 - Scale, smooth, adjust baseline drift, and resample motion profiles.
- **Multi-Axis Trajectory Planning:**
 - Full 6-DOF support (Translational: SI, AP, LR; Rotational: Roll, Pitch, Yaw).
 - Automated G-code generation optimized for Duet motion controllers.
- **Hardware Integration & Real-Time Control:**
 - Direct communication with Duet 3D motion platform controllers.
 - Real-time axis homing, position tracking, and manual jogging.
- **Triggered Imaging & Verification:**
 - Hardware synchronization with Geiger module for triggered imaging acquisition.
 - Live plotting and position error monitoring during motion execution.

1.2 User Guide Overview

This User Guide provides end-to-end instructions for setting up, operating, and troubleshooting TRACE:

1. **Installation:** Setting up the Python environment and launching TRACE.
2. **Hardware Setup:** Connecting the Duet controller, phantom stage, and Geiger trigger module.
3. **Files & Project Management:** Managing project directories and patient motion datasets.
4. **Motion Planning & G-code:** Creating analytical waveforms, editing patient curves, and exporting G-code.
5. **Phantom Control & Execution:** Homing, running motion trajectories, and triggering.
6. **Real-Time Status & Monitoring:** Live verification, logging, and performance metrics.
7. **Troubleshooting & Safety:** Resolving connection issues, limit switch stops, and safety protocols.



Download PDF Manual

When built with the `with-pdf` plugin, the complete documentation is available as a single downloadable PDF manual under `exports/TRACE_User_Guide.pdf`.

2. Installation & Software Setup

This page guides you through installing the required dependencies and launching the TRACE Graphical User Interface (GUI).

2.1 Prerequisites

Before installing TRACE, ensure you have installed:

- **Python 3.10 or later** (Conda or Miniconda environment recommended).
- **Git** (for repository cloning).

2.2 Installation Steps

Step 1: Clone the Repository

Clone the TRACE repository using Git or your preferred Git client (e.g., SourceTree, GitHub Desktop):

```
git clone https://github.com/YourUsername/TRACE.git
cd TRACE
```

Step 2: Create Conda Environment

TRACE provides an `environment.yml` file containing all required packages (PyQt / DearPyGui, NumPy, SciPy, Matplotlib, PySerial, etc.).

Create and activate the environment:

```
conda env create -f environment.yml
conda activate trace-env
```

Alternatively, if installing dependencies manually via `pip`:

```
pip install numpy scipy matplotlib pyqt5 pyserial requests
```

2.3 Configuration (`configuration.json`)

Check `configuration.json` in the root directory to verify hardware settings (IP address of Duet controller, default COM ports for Geiger module, default baud rates):

```
{
  "duet_ip": "192.168.1.100",
  "geiger_port": "COM3",
  "baud_rate": 115250,
  "default_output_dir": "./output"
}
```

2.4 Launching TRACE

To launch the graphical user interface, run `Launch_ImGUI.py`:

```
python Launch_ImGUI.py
```

 First Launch Verification

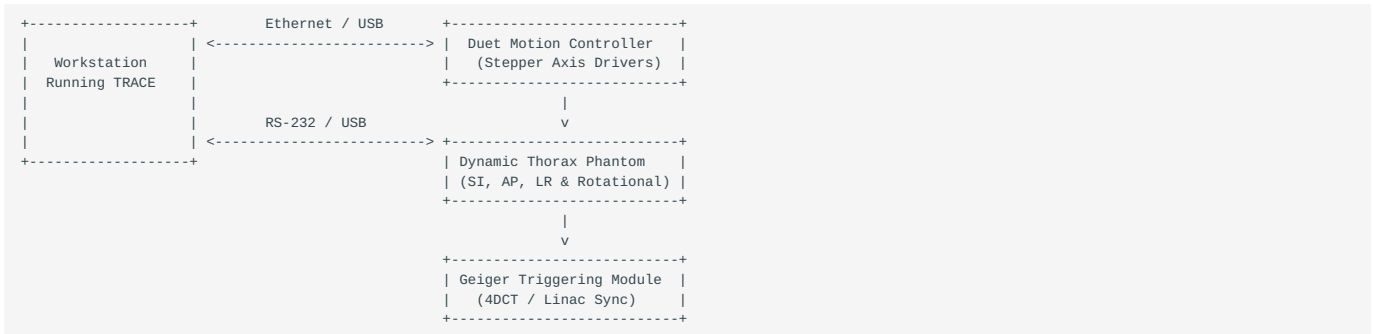
Upon successful launch, the TRACE interface window will open, displaying the navigation tabs (**Files, Planning, Control, Status**) and initializing the communication threads in the background.

3. Hardware Setup & Wiring

This guide details the physical hardware setup for connecting the dynamic anthropomorphic thorax phantom, the motion platform controller, and the triggering hardware to TRACE.

3.1 Hardware Architecture

The TRACE ecosystem consists of three main hardware components:



3.2 1. Duet Motion Controller Setup

TRACE communicates with a **Duet 3D motion controller** via HTTP REST API or Direct Serial Interface.

1. **Network Cable:** Connect an Ethernet cable from your workstation/switch to the Duet Ethernet port.
2. **IP Configuration:** Ensure the IP address matches `duet_ip` in `configuration.json` (Default: `192.168.1.100`).
3. **Axis Assignments:**
 - **Axis X:** Superior-Inferior (SI) translation.
 - **Axis Y:** Anterior-Posterior (AP) translation.
 - **Axis Z:** Left-Right (LR) translation.
 - **Axis U / V / W:** Rotational axes (Roll, Pitch, Yaw).

Emergency Stop Button

Always test the hardware E-STOP button wired to the Duet controller before running automated motion sequences.

3.3 2. Geiger Triggering Module Setup

The Geiger module enables synchronized X-ray imaging / 4DCT triggering during specific phases of phantom motion.

1. **Serial Connection:** Plug the USB-to-Serial adapter from the Geiger module into the workstation.
2. **Identify Port:** Note the assigned COM port (Windows: `COMx`, Linux/macOS: `/dev/ttyUSBx`).
3. **Trigger Output:** Connect the opto-isolated trigger output cable to the external trigger input of the imaging modality (Linac / CT scanner).

3.4 3. Power-On Checklist

Before launching TRACE:

- [] Verify 24V power supply connected to Duet controller.
- [] Confirm phantom motion stage is free from physical obstacles.
- [] Confirm Geiger module LED power indicator is green.
- [] Ping Duet IP (`ping 192.168.1.100`) from command prompt.

4. Chapter 1: Files & Project Management

The **Files Tab** in TRACE (`tab_files.py`) is the entry point for importing raw patient motion data, managing analytical waveform presets, and selecting output directories for generated G-code files.

4.1 Files Tab Overview

When you open the **Files** tab, you will find:

- **Project Directory Selector:** Choose where motion data and G-code outputs are saved.
- **Raw Data Import Panel:** Load CSV, TSV, DICOM, or log files containing respiratory motion traces.
- **Preset Library:** Select pre-defined standard motion templates (e.g., standard sinusoidal breathing, irregular breathing, high-frequency motion).

4.2 1. Importing Patient Respiratory Curves

TRACE supports importing patient-derived motion profiles captured by clinical motion tracking systems (such as RPM, Anzai, or Varian RGSC).

Supported File Formats

Format	Extensions	Description
CSV / TSV	<code>.csv</code> , <code>.tsv</code> , <code>.txt</code>	Columnar data containing time vs displacement columns.
RPM Logs	<code>.vms</code> , <code>.txt</code>	Varian Real-Time Position Management log files.
G-code	<code>.gcode</code> , <code>.g</code>	Pre-compiled Duet G-code trajectory files.

Step-by-Step Import Procedure

1. Click **Browse File** under the *Import Motion Data* panel.
2. Select your raw motion file.
3. Specify the time column index and motion axis column index.
4. Click **Parse File**. TRACE will render a preview plot of the imported waveform.

4.3 2. Managing Project Directory & Outputs

Outputs generated by TRACE (such as compiled `.gcode` files and trajectory logs) are stored in the selected Project Directory.

- **Set Default Directory:** Use **Select Directory** to choose your target folder.
- **File Naming Convention:**

```
<project_name>_<axis>_<waveform_type>.gcode
```

Example: `planned_motion_platform_SI_sin.gcode`

 Organizing Patient Datasets

We recommend creating a separate subfolder for each patient or phantom study inside your project directory to keep G-code and verification logs organized.

5. Chapter 2: Motion Planning & G-code Export

The **Planning Tab** (`tab_planning.py` & `fcn_plan`) is where motion profiles are designed, edited, mathematically transformed, and compiled into controller-ready G-code.

5.1 1. Creating Analytical Waveforms

TRACE allows you to generate synthetic mathematical motion curves to simulate standard breathing patterns or mechanical test signals.

Available Analytical Waveform Types

1. **Sinusoidal Curve:** $z(t) = A \cdot \sin\left(\frac{2\pi}{T} t + \phi\right) + z_0$
 2. A : Amplitude (mm)
 3. T : Breathing period (seconds)
 4. ϕ : Phase offset (rad)
 5. z_0 : Baseline offset (mm)
 6. **Modified $\cos^4$ Breathing Curve:** Simulates realistic asymmetry between inspiration and expiration phases: $z(t) = A \cdot \cos^4\left(\frac{\pi}{T} t\right) + z_0$
 7. **Trapezoidal / Step Motion:** Used for step response tests and static phantom offset calibrations.
-

5.2 2. Editing Patient-Derived Motion Curves

Raw clinical respiratory traces often contain baseline drift, high-frequency noise, or cardiac artifacts. TRACE provides interactive tools in `fcn_edit.py` to refine these profiles:

- **Baseline Correction:** Removes gradual baseline elevation drift over time.
 - **Amplitude Scaling:** Rescales motion to match specific phantom limits (e.g., max 30 mm SI displacement).
 - **Smoothing / Filtering:** Applies Savitzky-Golay or low-pass Butterworth filtering to eliminate signal noise.
 - **Resampling:** Interpolates motion data to match the Duet motor update frequency (typically 50 Hz or 100 Hz).
-

5.3 3. Multi-Axis Preview

Before exporting G-code, preview the 6-DOF motion components in the interactive plot panel:

- **SI Axis (Z-motion):** Primary Superior-Inferior displacement.
 - **AP Axis (Y-motion):** Anterior-Posterior displacement.
 - **LR Axis (X-motion):** Left-Right displacement.
 - **Rotational Axes (Roll, Pitch, Yaw):** Rotational tilting.
-

5.4 4. Exporting G-Code for Duet Controller

Once you are satisfied with the trajectory:

1. Select target motor acceleration and velocity limits.
2. Click **Generate G-Code** (`fcn_gcode.py`).
3. TRACE compiles linear/curved segment commands:

```
G90 ; Absolute positioning mode
G1 X12.4 Y2.1 Z-5.0 F1200 ; Linear move with feedrate 1200 mm/min
G1 X12.8 Y2.3 Z-4.8 F1250
```

4. Click **Save G-Code** to store the `.gcode` file into your project output directory.

6. Chapter 3: Phantom Control & Execution

The **Control Tab** (`tab_control.py` & `fcn_control.py`) provides direct control over the dynamic thorax phantom hardware, motor initialization, manual jogging, G-code execution, and trigger synchronization.

6.1 1. Connecting to Hardware

1. Navigate to the **Control** tab.
 2. Select connection mode: **Ethernet (REST API)** or **Serial (COM)**.
 3. Click **Connect Controller**.
 4. Check connection status indicator:
 - **Connected:** IP / COM port open and responding.
 - **Disconnected:** Verify wiring or IP settings in `configuration.json`.
-

6.2 2. Homing & Zeroing Axes

Before executing any motion trajectory, the physical platform **MUST** be homed to establish mechanical zero reference points.

1. Ensure the platform path is clear.
2. Click **Home All Axes** (`\(G28\)`).
3. Wait for all axis status indicators to turn green.
4. Verify position readout shows `(0.00, 0.00, 0.00)`.

Axis Safety Limits

Never attempt to run G-code without homing first. Running motion profiles from un-homed positions can cause mechanical collision against endstops.

6.3 3. Manual Jogging

Use the manual jog panel to fine-tune phantom position or perform pre-experiment align checks:

- **Step Sizes:** `\(0.1\text{mm}, 1.0\text{mm}, 5.0\text{mm}, 10.0\text{mm}\)`.
 - **Directions:** `\(pm X, pm Y, pm Z, pm \text{Roll}, pm \text{Pitch}, pm \text{Yaw}\)`.
 - **Feedrate Control:** Adjustable speed slider (`\(10 - 3000\text{mm/min}\)`).
-

6.4 4. Running Motion Trajectories

To execute a planned G-code motion file:

1. Click **Select G-Code File** and pick your compiled trajectory.
 2. Select execution mode:
 - **Single Cycle:** Plays motion sequence once and returns to home.
 - **Continuous Loop:** Repeats motion sequence until manually stopped.
 3. Click **Start Motion**.
 4. Use **Pause Motion** ($\backslash(M600\backslash)$) or **Emergency Stop** ($\backslash(M112\backslash)$) at any time to halt movement.
-

6.5 5. Geiger Triggering Configuration

To enable imaging/gating triggers during specific phases of the trajectory:

1. Toggle **Enable Geiger Trigger Sync**.
2. Set **Phase Trigger Point** (e.g., peak inspiration at $\backslash(90^\circ\backslash)$ or end expiration at $\backslash(270^\circ\backslash)$).
3. Set **Pulse Duration** (e.g., $\backslash(50\backslash\text{ms}\backslash)$).
4. Click **Test Trigger Signal** to verify optocoupler pulse output on the Geiger hardware module.

7. Chapter 4: Real-Time Status & Monitoring

The **Status Tab** (`tab_status.py`, `fcn_duet.py`, `fcn_plotting.py`) monitors motion execution in real time, displays live feedback plots, and logs trigger events for post-experiment validation.

7.1 1. Live Motion Tracking Plots

During motion execution, TRACE renders real-time multi-axis tracking plots:

- **Planned Trajectory (Blue)**: Target position requested by G-code.
 - **Actual Feedback Position (Red)**: Encoder / motor feedback reported by controller.
 - **Position Error (Green)**: Difference $\Delta z(t) = z_{\text{actual}}(t) - z_{\text{planned}}(t)$.
-

7.2 2. Dynamic Performance Indicators

The telemetry panel displays:

- **Current Axis Positions**: Live coordinates $((X, Y, Z, U, V, W))$.
 - **Actual Speed / Velocity**: Real-time motion speed in (mm/s) .
 - **Buffer State**: Controller G-code command buffer fill level (prevents buffer underruns during high-frequency motion).
 - **Trigger Event Counter**: Total number of Geiger trigger pulses issued during the session.
-

7.3 3. Logging & Data Export

TRACE automatically records execution logs for motion quality assurance (QA):

- **Save Log File**: Exports execution history to CSV format (`motion_log_YYYYMMDD_HHMMSS.csv`).
- **QA Metrics**:
 - **Root Mean Square Error (RMSE)** across the entire trajectory:
$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N \left(z_{\text{actual}}(i) - z_{\text{planned}}(i) \right)^2}$$
 - **Max Peak Position Error**.
 - **Trigger Jitter**: Timing variation between planned phase and actual electrical pulse.

8. Troubleshooting & Safety Guide

This guide covers common operational issues, hardware error messages, recovery procedures, and safety protocols when running TRACE with the dynamic phantom.

8.1 Hardware Connection Issues

1. "Duet Controller Disconnected / Timeout"

Symptoms: Red status indicator, commands fail to send.

Solutions: - Verify Ethernet cable connection between workstation and Duet. - Test network connectivity:

```
ping 192.168.1.100
```

- Open browser and navigate to `http://192.168.1.100` to confirm Duet Web Control (DWC) is responding. - Verify IP address matches `duet_ip` in `configuration.json`.

2. "Geiger Serial Port COMx Not Found"

Symptoms: Cannot toggle trigger synchronization or test pulses.

Solutions: - Unplug USB-to-Serial cable and re-insert. - Open Device Manager (Windows) or run `ls /dev/tty*` (Linux) to check assigned port number. - Update `geiger_port` in `configuration.json` and restart TRACE.

8.2 Motion & Limit Switch Errors

3. "Axis Hit Limit Switch / Endstop Triggered"

Symptoms: Emergency stop activated, movement freezes mid-trajectory.

Solutions: 1. Check physical platform position for mechanical obstruction or extreme travel. 2. Click **Reset Controller Alarm**. 3. Use manual jog controls in reverse direction at low speed (0.1 mm/s) to backed off switch. 4. Verify motion trajectory limits in **Planning Tab** before re-running.

4. "Buffer Underrun / Stuttering Motion"

Symptoms: Phantom movement stutters or pauses intermittently.

Solutions: - Reduce point density / increase step interval in **Planning Tab** during G-code generation. - Close CPU-intensive background applications. - Switch connection mode from Serial to Ethernet REST API for higher throughput.

8.3 Safety Protocols



Laboratory Safety Guidelines

- **Never place hands or tools inside the phantom motion stage while power is applied.**
- Keep the hardware Emergency Stop button within easy reach during all automated trajectory runs.
- Inspect motor belts and mechanical fasteners weekly for wear or tension loss.